Advanced Gold Nanorod-Based Theranostics: A Multimodal Approach to Combat Pancreatic Ductal Adenocarcinoma

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Pancreatic ductal adenocarcinoma (PDAC) is notably one of the most hypoxic and treatment-resistant cancer types [1]. To tackle this issue, we have developed AuNRs-GEM, an innovative theranostic agent aimed at PDAC. This agent integrates the chemotherapeutic power of gemcitabine with thermal therapy induced by near-infrared (NIR) light absorption.

The dual therapeutic strategy capitalizes on the advantages of both chemotherapy and localized hyperthermia. Gemcitabine, a standard chemotherapeutic drug [2], is effective against PDAC, while hyperthermia, induced by the NIR absorption of gold nanorods (AuNRs) [3], improves drug delivery by enhancing vessel dilation and modifies tumor oxygen partial pressure (pO_2). This multimodal treatment approach seeks to overcome the inherent resistance of hypoxic tumors to conventional therapies.

Furthermore, AuNRs-GEM facilitates high-resolution imaging through photoacoustic imaging and computed tomography (CT), offering valuable insights into tumor morphology and treatment efficacy. By combining these diagnostic and therapeutic techniques, we aim to enhance the overall treatment outcomes for PDAC patients.

The AuNRs-GEM nanohybrid was synthesized through amide bond formation between carboxylated gold nanorods and gemcitabine using the water-soluble carbodiimide method (EDCI-NHS). Characterization of the nanohybrid was performed using UV-VIS, FT-IR, and XPS spectroscopies. The concentration of AuNRs-GEM was determined using RP-HPLC and TXRF techniques. The morphology and structure of the gold nanomaterial were analyzed by TEM and SEM, followed by DLS measurements and zeta potential assessment.

In vitro testing of the gold theranostic agent was conducted on mouse and human PDAC cell lines (Panc1, Pan_O2, AsPc1). Metabolic activity and cell number were measured after exposure to AuNRs-GEM and light under both hypoxic and normoxic conditions. A PDAC model in C57BL/6J mice was established using the Pan_O2 cell line. Tumor oxygenation was measured using electron paramagnetic resonance (EPR) spectroscopy, with Jiva-25 or Bruker E540L instruments and trityl OX071 or Oxychip as the spin probe. Tumor anatomy and vascular structure were evaluated using ultrasound, including B-mode and Power Doppler with Vevo F2 from FujiFilm VisualSonics.

The therapeutic regimen involved administering AuNRs-GEM (gold nanorods loaded with gemcitabine) combined with hyperthermia using near-infrared light (~808 nm). The treatment included six doses of AuNRs (approximately 1 µg/ml) and GEM (approximately 45 mg/kg BW), along with near-infrared heating (approximately 3 x 1.5 min during 12 min) administered every 72 hours. All experiments were approved by the Local Ethics Committee (no. 151/2022 and no. 250/2020).

The developed compound shows promise in enhancing the efficacy of treatments for unresectable PDAC in the future. Tumor oxygen partial pressure (pO_2) can be predictive of therapy outcomes.

References

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